

Solving Problems Involving Trapezoids and Kites

Jonathan R. Bacolod

A **trapezoid** is a quadrilateral with one pair of parallel sides called the **bases**.

Median of a Trapezoid

The **median (midsegment)** of a trapezoid is parallel to the bases and its length is

$$m = \frac{b_1 + b_2}{2}.$$

Area of a Trapezoid

The area of a trapezoid is

$$A = \frac{1}{2}(b_1 + b_2)h.$$

Isosceles Trapezoid

An **isosceles trapezoid** has congruent legs, congruent base angles, and congruent diagonals.

Kite

A **kite** is a quadrilateral with two pairs of adjacent congruent sides.

Diagonals of a Kite

The diagonals of a **kite** are perpendicular, and one diagonal bisects the other.

$$d_1 \perp d_2$$

Area of a Kite

The area of a kite is

$$A = \frac{1}{2}d_1d_2.$$

Solving Problems

Use the properties of **trapezoids** and **kites** to solve real-life and mathematical problems involving lengths, angles, perimeters, and areas.

Examples

Example 1

A trapezoid has bases of 12 cm and 20 cm with a height of 8 cm. Find its area.

$$A = \frac{1}{2}(b_1 + b_2)h$$

$$A = \frac{1}{2}(12 + 20)(8)$$

$$A = 128$$

The area is $\boxed{128 \text{ cm}^2}$.

Example 2

An isosceles trapezoid has base angles measuring $(3x + 10)^\circ$ and $(5x - 30)^\circ$. Find x .
The base angles are congruent.

$$3x + 10 = 5x - 30$$

$$40 = 2x$$

$$x = 20$$

Therefore, $\boxed{x = 20}$.

Example 3

One diagonal of a kite measures 24 cm. If its area is 180 cm^2 , find the other diagonal.

$$A = \frac{1}{2}d_1d_2$$

$$180 = \frac{1}{2}(24)d_2$$

$$180 = 12d_2$$

$$d_2 = 15$$

The other diagonal is 15 cm.

Example 4

The median of a trapezoid is 21 cm. One base is 16 cm. Find the other base.

$$m = \frac{b_1 + b_2}{2}$$

$$21 = \frac{16 + b}{2}$$

$$42 = 16 + b$$

$$b = 26$$

The other base is 26 cm.

Example 5

A farmer owns a trapezoidal field with bases of 40 m and 70 m and a height of 25 m. Find the area of the field.

$$A = \frac{1}{2}(b_1 + b_2)h$$

$$A = \frac{1}{2}(40 + 70)(25)$$

$$A = 1375$$

The area is 1375 m².

Practice Exercises

Practice 1

A trapezoid has bases of 12 cm and 20 cm with a height of 8 cm. Find its area.

$$A = \frac{1}{2}(b_1 + b_2)h$$

Practice 2

An isosceles trapezoid has base angles measuring $(3x + 10)^\circ$ and $(5x - 30)^\circ$. Find x .

$$3x + 10 = 5x - 30$$

Practice 3

One diagonal of a kite measures 24 cm. If its area is 180 cm^2 , find the length of the other diagonal.

$$A = \frac{1}{2}d_1d_2$$

Practice 4

The median of a trapezoid is 21 cm. One base is 16 cm. Find the other base.

$$m = \frac{b_1 + b_2}{2}$$

Practice 5

A farmer owns a trapezoidal field with bases of 40 m and 70 m and a height of 25 m. Find the area of the field.

$$A = \frac{1}{2}(b_1 + b_2)h$$

Activities

Activity 1

The bases of a trapezoid are 18 cm and 30 cm.
Find the length of its median.

$$m = \frac{18 + 30}{2}$$

Activity 2

A kite has diagonals measuring 16 cm and 22 cm. Find its area.

$$A = \frac{1}{2}d_1d_2$$

Activity 3

A trapezoid has bases of 24 m and 36 m with a height of 18 m. Find its area.

$$A = \frac{1}{2}(b_1 + b_2)h$$

Activity 4

A kite has adjacent sides measuring 15 cm, 15 cm, 12 cm, and 12 cm. Find its perimeter.

$$P = 15 + 15 + 12 + 12$$

Activity 5

A kite-shaped playground has diagonals measuring 40 m and 32 m. If rubber flooring costs ₱220 per square meter, how much will it cost to cover the playground?

$$A = \frac{1}{2}(40)(32)$$

Thank you!